

Charlotte Wayment

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Location: Oxford, UK

EDUCATION

DPhil, Autonomous and Intelligent Machines and Systems (AIMS CDT) Oct 2025 – Oct 2029
University of Oxford, UK

- **Scholarship:** Trinity College Montgomery Scholarship (2025-29); Trinity College Marconi-Moseley Prize for Science in the Service of Humanity; AIMS CDT EPSRC Studentship (2025-29)
- **Current Research: Reliability-aware assistive device control under signal degradation and failure:**
 - Developing nonparametrically calibrated, failure-aligned reliability measures for multimodal biosignal fusion.
 - Investigating online calibration under non-stationarity and decision-making under context-dependent failure costs.

MSc, Data Science Sep 2024 – Sep 2025
Durham University, UK

- **Classification:** Graduated **First in Year** with a **Distinction** and a grade average of 86%.
- **Thesis (89%):** "Forecasting using GAT-LSTM: Capturing Spatio-Temporal Dynamics in Sparsely Gauged, Noisy Sensor Networks" supervised by Dr Julia Knapp.
- **Skills:** Modelling Noisy, Sparse, and Unreliable Sensor Data; GNNs; Efficient & Reproducible ML Pipelines; HPC

MA (Hons), Architecture Sep 2020 – Jul 2024
University of Edinburgh, UK

- **Classification: First Class Degree with Honours;** placing **First in Year** across all four years of study.
- **Thesis (82%):** "Urban-Scale Modelling for a Net-Zero Scotland: How Graph-Based Representations Can Be Leveraged to Inform Retrofitting in Edinburgh" supervised by Dr Alessia Calafiore.

SKILLS

- **General:** Uncertainty Quantification & Nonparametric Calibration; Multimodal & Biosignal Modelling; Machine Learning & Neural Networks (PyTorch); Structured Modelling of Constrained Systems; Numerical Methods & Scientific Computing; Reproducible ML Pipelines; Python Scientific Stack (inc. SciPy, Pandas); HPC (Slurm); Linux

EXPERIENCE

Python Software & ML Engineer Part-Time (Full-Time during Holidays) 2021 – 2025
Wolfwyse Ltd.

- Designed and deployed a production-grade scheduling algorithm, reducing multi-day planning tasks to minutes by formalising complex operational constraints, dependencies, and resource flows into computational models.
- Developed reproducible optimisation pipelines combining constraint-based methods and evolutionary algorithms, integrated into internal APIs and deployed in safety- and reliability-critical production environments with automated feasibility, constraint, and regression checks - informing later work in safety-critical ML systems
- Collaborated with engineers to translate networked operational processes into tractable optimisation problems, enabling robust scenario evaluation under real-world manufacturing conditions.

Skills: Python; Version Control; Evolutionary Algorithms; Constraint-Based Optimisation; Reproducible Pipelines; Unit Testing

Simulations Engineer Academic 2023 – 2024
The University of Edinburgh Hyperloop Team (HYPED)

- Supported an interdisciplinary team to develop a working Hyperloop system prototype for European Hyperloop Week.
- Performed multi-scale CFD and solid mechanics simulations using COMSOL Multiphysics to inform structural design decisions and identify high-risk failure points in a complex electromechanical system.
- Supported rapid simulation-informed prototype iteration to reduce manufacturing waste by more than 30% during R&D (compared to 2022-23 development cycle).

Skills: Simulation & Numerical Modelling (COMSOL Multiphysics, CFD); Prototype R&D; Physics-Constrained Optimisation

Project Engineer & Technical Support Full-time 2018 – 2020; Contractor 2021 – 2023
J P Hildreth Ltd. Engineering Consultancy

- Developed quantitative models of spatially and operationally constrained systems to optimise production layouts and workflows for highly-regulated food and pharmaceutical manufacturing environments (inc. GSK, Mars Chocolate).
- Translated facility-specific manufacturing conditions into priority-driven future production scenarios (inc. CAD-based spatial models), communicating scenario trade-offs to technical and non-technical clients.

Skills: CAD Design; Applied Modelling & Optimisation; Technical Reporting; Client-Facing Technical Communication

AWARDS

- **Marconi-Moseley Prize:** Awarded annually to an early-career researcher at Trinity College, Oxford. **Winner 2026**
- **The G.A. Curry MSc Prize:** Awarded for the best MSc Thesis in an earth sciences application in the UK. **Winner 2026**
- **Master of Data Science Award:** Best performance across all Data Science Masters programme streams. **Winner 2025**
- **The ESALA Architecture Awards & Helen A. Rose:** Awarded to the highest performing student in years 1, 2, & 3; and the highest achieving graduating student across BA/MA (Hons) Architecture degrees at Edinburgh. **Winner 2021 – 2024**
- **The Nancy Rosemary Henderson & David Gardinier Hardie Technology Prizes:** Awarded to the best performing student in engineering and technology courses across undergraduate years one and two respectively. **Winner 2021, 2022**